

ARTICLE



Comparing efficacy of charged-particle therapy with brachytherapy in treatment of uveal melanoma

Yu-Hsuan Tseng¹, Chia-An Hsu² and Yu-Bai Chou^{2,3,4}✉

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BACKGROUND: Uveal melanoma (UM) is the most common primary ocular tumour in adults. The most used eye-preserving treatments are charged-particle therapy (CPT) and brachytherapy. We performed a systematic review and meta-analysis to compare efficacies and complications of these two radiotherapies.

METHODS: We searched EMBASE, PubMed, MEDLINE, and the Cochrane Library from January 2012 to December 2022. Two independent reviewers identified controlled studies comparing outcomes of CPT versus brachytherapy. Case series that utilize either treatment modality were also reviewed.

RESULTS: One hundred fifty studies met the eligibility criteria, including 2 randomized control trials, 5 controlled cohort studies, and 143 case series studies. We found significant reduction in local recurrence rate among patients treated with CPT compared to brachytherapy (Odds ratio[OR] 0.38, 95% Confidence interval [CI] 0.24–0.60, $p < 0.01$). Analysis also showed a trend of increased risks of secondary glaucoma after CPT. No statistically significant differences were found in analyzing risks of mortality, enucleation, and cataract.

CONCLUSIONS: Our study suggested no difference in mortality, enucleation rate and cataract formation rate comparing the two treatments. Lower local recurrence rate and possibly higher secondary glaucoma incidence was noted among patients treated with CPT. Nevertheless, the overall level of evidence is limited, and further high-quality studies are necessary to provide a more definitive conclusion.

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INTRODUCTION

Uveal melanoma (UM) is the most common primary intraocular tumour in adults [1] with high mortality and morbidity rate. Traditionally, surgical enucleation was considered the principal treatment for uveal melanoma. Nevertheless, this approach results in not only vision loss but also cosmetic problem. Therefore, there has been a growing interest in exploring eye-preserving treatments, such as radiotherapy. The mainstream of primary radiotherapy for UM can be subdivided into internal radiation therapy (e.g. brachytherapy) and external radiation therapy (e.g. charged-particle therapy, CPT).

Brachytherapy is a form of radiation therapy that involves placing a radioactive source directly inside or adjacent to the targeted treatment area and kept in place for 2–7 days depending on the isotope used. The milestone randomized control trial, the Collaborative Ocular Melanoma Study (COMS) [2, 3], revealed no difference in 5-year all-cause mortality for large- and medium-sized choroidal melanomas comparing enucleation and brachytherapy. Brachytherapy had since become the standard treatment for UM, and there had been growing trend in developing eye-preserving therapies. The two most commonly utilized radioisotope sources are ruthenium-106 and iodine-125. Both isotopes demonstrated comparable

efficacy in terms of mortality, enucleation rate, and local control rate [4].

CPT is a type of external beam radiation that utilizes charged particles as radiation sources, including protons, carbon ions and helium ions. Initially, CPT therapy for uveal melanomas were done in the United States with protons in Boston and with helium ions in Berkeley [5]. Clinical trials for carbon ion radiotherapy were conducted a few years later in Japan and Germany [6]. In contemporary cancer treatment, protons and carbon ions are the most commonly employed charged particles [7]. Helium ions demonstrate comparable physical characteristics to other heavy ions, such as carbon ions, and are increasingly gaining attention in scientific research [5]. In contrast to conventional radiotherapy, charged particles can be precisely directed and deliver maximum energy at specific depths within the body, known as the Bragg peak, while emitting a minimal dose of energy before and beyond the Bragg peak. This allows more precise targeting and treatment of the tumour. However, the major limitations of CPT are the high expenses and limited availability, with specialized expertise and technology restricted to a handful of centres worldwide.

At present, there is still lack of definite clinical guidance regarding the selection of optimal treatment modalities, although some rationale for the choice can be listed. Brachytherapy is

¹Department of Medical Education, Taipei Veterans General Hospital, Taipei 11217, Taiwan. ²Department of Ophthalmology, Taipei Veterans General Hospital, Taipei 11217, Taiwan. ³School of Medicine, National Yang Ming Chiao Tung University, Taipei, Taiwan. ⁴Institute of Public Health, National Yang Ming Chiao Tung University, Taipei, Taiwan. ✉email: suddenonset@gmail.com

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indicated in small- and medium-sized tumours, and contra-indicated in large tumours and tumours within 2 mm from the optic disc. CPT may be considered in larger tumours with a height of more than 5 mm, tumours with a narrow base, tumours close to the optic nerve, and when there is ciliary body involvement [8, 9]. Numerous studies have investigated the efficacy and outcome of these two treatment modalities, including mortality, eye-preservation rate, metastatic rate, local recurrence, and post-treatment complications. However, the majority of these studies involve patients recruited from a single institution treated by either CPT or brachytherapy, and only few studies comparing the two eye-preserving radiotherapies with limited resources. Nowadays, there are emerging interest in the use of CPT in treating various types of tumour [10, 11], including UM, and more studies focused on this field to evaluate the efficacy, outcome, and complications by different study design, including head-to-head comparison studies of different treatment modalities [12–15]. However, the results were controversial, and currently there are no definitive clinical guidelines available to guide treatment selection. Therefore, more literature should be reviewed and summarized for updated meta-analysis to provide robust evidences.

The objective of the present study is to provide a systematic review and meta-analysis of more recent research studies comparing the efficacy and complication rates of eye-preserving radiotherapy in patients with UM.

MATERIALS AND METHODS

We conducted a systematic review and meta-analysis on the topic of UM treated with CPT compared with brachytherapy. This study was registered with PROSPERO (ID:399075.) The methodology and reporting of this study are consistent with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

Search strategy

We searched EMBASE, PubMed, MEDLINE, and the Cochrane Library from January 2012 to December 2022, using appropriate database terms. These terms included “helium ion,” “carbon ion,” “proton beam,” “charged particles,” “brachytherapy,” “iodine-125,” “ruthenium-106,” “plaque therapy,” “eye melanoma,” “ocular melanoma,” “uveal melanoma,” “choroidal melanoma,” “ciliary melanoma,” and MeSH terms “proton therapy,” “radiotherapy,” “brachytherapy,” and “uveal neoplasms,” along with appropriate Boolean algebra. We also evaluated non-English studies, which were translated through proper measures, for eligibility. In addition, we screened clinical reviews to identify relevant studies. We identified relevant studies published before 2012 from 5 systematic reviews. A detailed search strategy is available in the supplementary information.

Eligibility criteria

Studies were eligible if they used charged-particle therapy or brachytherapy as the primary treatment for uveal melanomas. The former included helium ions, carbon ions, and proton beam therapy, while the latter included iodine-125 and ruthenium-106 plaque therapy. Studies were excluded if these radiotherapies were used as neoadjuvant therapy, adjuvant therapy, or treatment for recurrent melanomas. Long-term follow-up studies that only included statistics at 15 years or more were also excluded. The search results were reviewed independently by two reviewers according to the inclusion criteria. In cases of discrepancies, a third author was consulted.

Data collection and risk-of-bias assessment

We collected results of patients treated with CPT (the CPT group) and brachytherapy (the brachytherapy group) respectively. The following parameters were extracted independently by two authors from the studies: publication year, follow-up time, number of patients, country, duration, age of patients, radiation dose, tumour size, tumour location, 5-year disease-specific mortality, 5-year local recurrence rate, total enucleation rate, and complications including radiation-induced cataract and secondary glaucoma. Two authors performed the assessment of study

quality independently. For randomized control trials (RCTs), we applied the Cochrane risk-of-bias tool [16] to assess the methodological quality. The Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) assessment tool [17] by Cochrane Collaboration was utilized for cohort studies, and the Joanna Briggs Institute (JBI) checklists [18] was used for case series studies.

Data synthesis and analysis

Meta-analysis was conducted utilizing R version 4.1.3 (R Foundation for Statistical Computing, Vienna, Austria) and the “meta” and “metafor” package. Outcomes of the treatment, including death, total enucleation rate, local recurrence rate and incidences of complications were obtained from comparative studies expressed in proportions and were fit into a meta-regression model based on treatment method.

We also analyzed the outcomes of charged-particle therapy and brachytherapy from case series. Odds ratio and 95% confidence intervals were obtained using a meta-regression model based on treatment method to compare the outcomes of the two treatments. Heterogeneity across individual studies were measured using τ^2 index, I^2 index, H^2 statistic and Cochran's Q test.

Sensitivity analysis

We utilized the “dmetar” package in R to analyze the odds ratio of outcomes between the CPT group and brachytherapy groups in each controlled study and summarized the results using Mantel-Haenszel method. The sensitivity analysis provided an alternate approach to assessing the results.

RESULTS

Study inclusion and demographics

After searching the three databases, a reference list of 2244 records was identified, and 1102 records were eligible for title and abstract screening after removing duplicates. Out of the 1102 records screened, 306 were retrieved for full-text screening. Among these, 163 articles were excluded due to the absence of interested outcome measurement. The reviewers eventually agreed to include 143 records for quantitative synthesis, according to the eligibility criteria. These included 2 randomized control trials [15, 19], 5 controlled cohort studies [12–14, 20, 21], and 136 case series studies [14, 22–146]. The search flow is presented in Fig. 1.

Demographics

Descriptive characteristics of the studies included were summarized in Table S1. A total of 48,944 patients with uveal melanoma were included in this review. Among them, 3925 patients were included in randomized control trial or cohort studies, and 45,019 patients were included in case series. Of the 3925 patients, 2364 patients received brachytherapy, and 1561 patients received CPT.

Overall efficacy: cohort studies analysis

Table 1 and Fig. 2a displayed the odds ratio (OR) and 95% confidence interval (CI) predicted from cohort studies, comparing CPT and brachytherapy in death, enucleation, local tumour recurrence, cataract, and secondary glaucoma. The mortality analysis comprised five studies, and no statistically significant differences were observed between the CPT group and the brachytherapy group concerning mortality (OR 1.49, 95% CI 0.38–5.88, $p = 0.57$). For the analysis on enucleation rate, four studies were included, and no significant reduction was found (OR 0.97, 95% CI 0.30–3.08, $p = 0.96$) in the CPT group. The present study found a significant reduction in the local tumour recurrence rate among patients treated with CPT (OR 0.38, 95% CI 0.24–0.60, $p < 0.01$), including five studies. Two studies and four studies were respectively included in the analysis of treatment complications for cataract and secondary glaucoma, and no statistically significant differences were found comparing the two treatment modalities (cataract: OR 0.71, 95% CI 0.06–8.20,

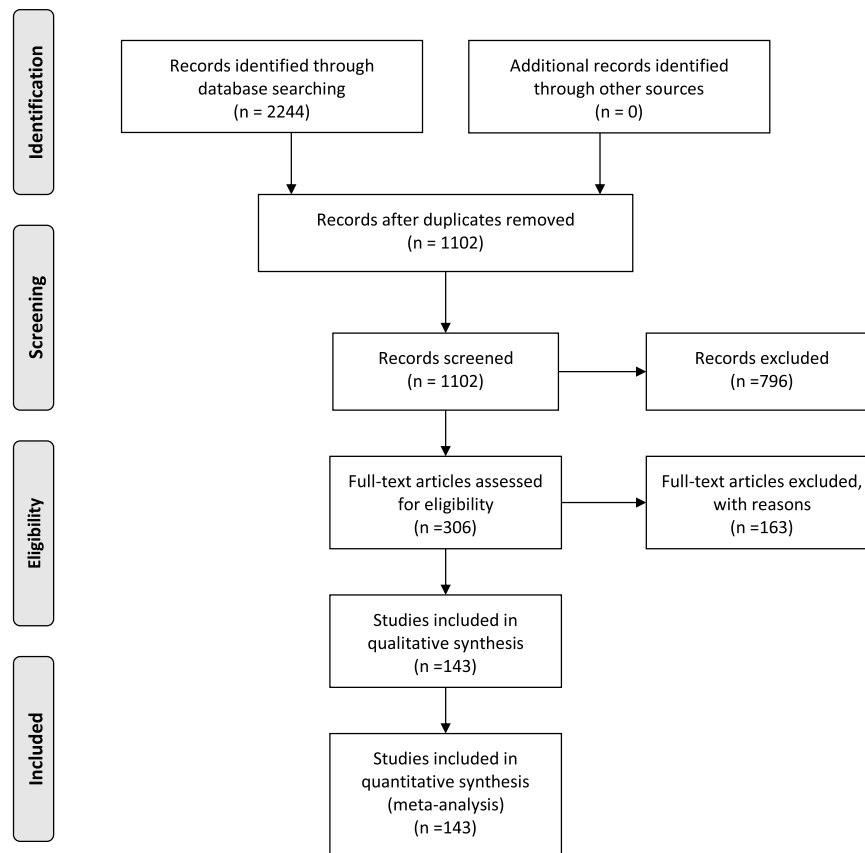


Fig. 1 Flow diagram. The Preferred reporting Items for systematic reviews and meta-analysis flow diagram.

Table 1. Meta-regression results of efficacy retrieved from cohort studies comparing CPT and brachytherapy.

	No. of patients	No. of studies included	OR	95% CI		<i>p</i> value
Death	2161	4	1.49	0.38	5.88	0.57
Enucleation	937	3	0.97	0.3	3.08	0.96
Local tumour recurrence	2545	4	0.38	0.24	0.6	<0.01
Cataract	340	2	0.71	0.06	8.2	0.79
Secondary glaucoma	937	3	2.68	0.53	13.47	0.23

$p = 0.79$; secondary glaucoma: OR 2.68, 95% CI 0.53–13.47, $p = 0.23$).

Overall efficacy: case series analysis

Table 2 and Fig. 2b presented the case series pooling results of the predicted odds ratio (OR) and 95% confidence interval (CI) of death, enucleation, local tumour recurrence, cataract, and secondary glaucoma, comparing CPT to brachytherapy. The number of studies and patients included was displayed in the table. The study found no statistically significant difference in mortality (OR 1.32, 95% CI 0.99–1.77, $p = 0.06$) between the two groups. There was also no statistical significance in the difference of enucleation rate (OR 1.16, 95% CI 0.87–1.54, $p = 0.31$) among the two groups. A significant reduction in local tumour recurrence (OR 0.54, 95% CI 0.40–0.72, $p < 0.01$) was found in the CPT group. The study showed no significant difference in cataract between the two groups (OR 1.59, 95% CI 0.87–2.93, $p = 0.13$). However, a statistically significant higher rate of secondary glaucoma (OR 1.81, 95% CI 1.24–2.63, $p < 0.01$) was found among those treated with CPT.

Quality assessment and heterogeneity

We applied the Cochrane risk-of-bias tool for assessing RCTs. For non-randomized, comparative cohort studies, The ROBINS-I assessment tool was used for qualitative assessment, and the results were shown in Fig. S1. Most of the studies had some concerns in overall bias and are graded moderate to serious risk of bias.

Sensitivity analysis

We conducted a sensitivity analysis by summarizing the odds ratio of outcomes between the CPT group and brachytherapy groups in each controlled study. The results were listed in Table 3. We did not find significant differences in risks of mortality (OR 1.38, 95% CI 0.53–3.58, $p = 0.4$) and enucleation (OR 0.85, 95% CI 0.24–3.03, $p = 0.71$) between the two treatment modalities, which was consistent with our main findings. No statistically significant reduction of local recurrence rate was found in the CPT group, but the analysis revealed a trend favouring CPT (OR 0.27, 95% CI 0.06–1.19, $p = 0.07$). The sensitivity analysis yielded similar results to our main analysis.

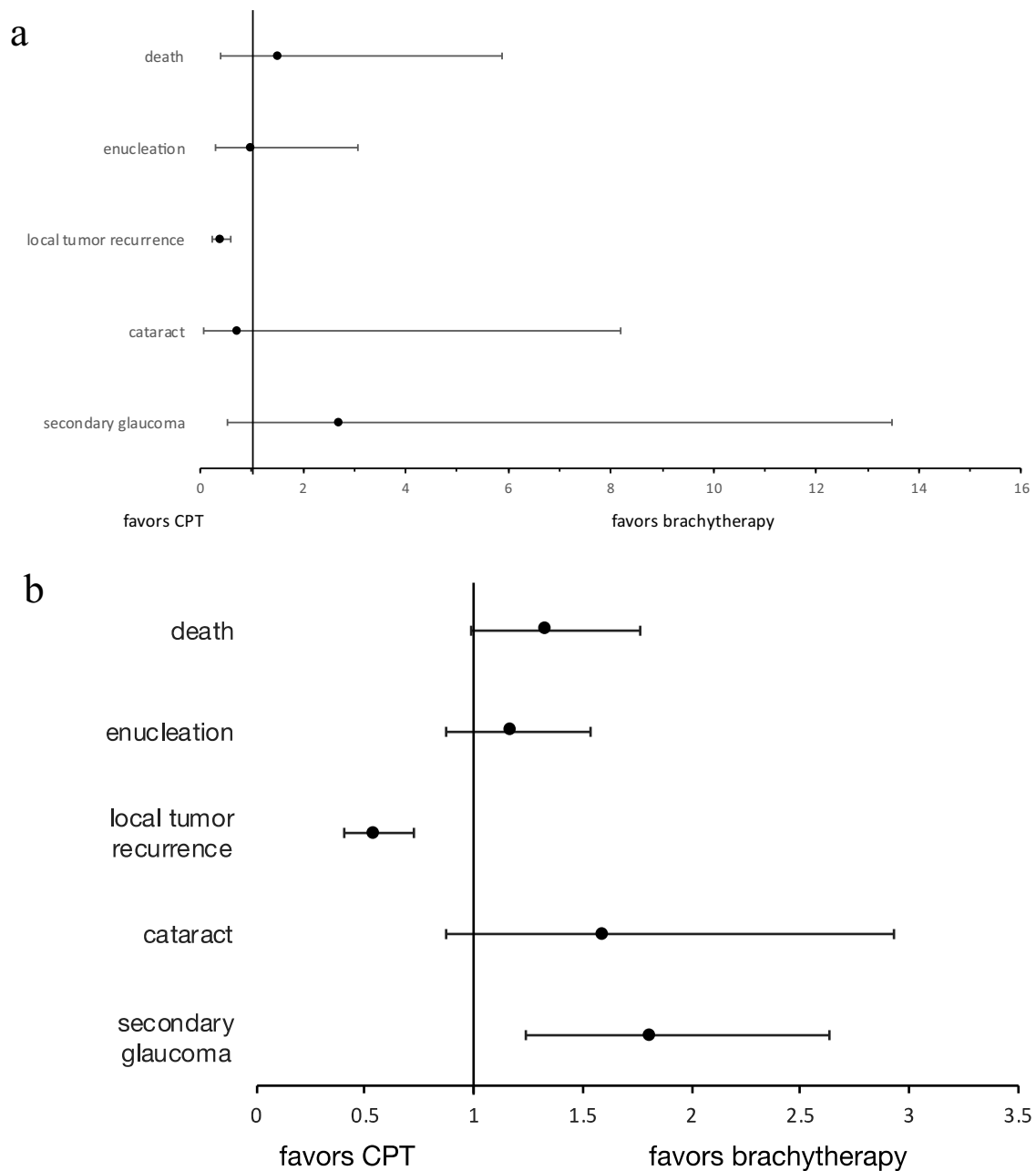


Fig. 2 OR and 95% CI of efficacy comparing CPT to brachytherapy. **a** OR and 95% CI of efficacy retrieved from cohort studies comparing CPT to brachytherapy. Significant reduction in the local tumour recurrence rate were found among patients treated with CPT (OR 0.38, 95% CI 0.24–0.60, $p < 0.01$), but there were no statistically significant differences observed between the CPT group and the brachytherapy group concerning mortality, enucleation rate, cataract and secondary glaucoma rates. **b** OR and 95% CI of efficacy retrieved from case series comparing CPT to brachytherapy. A significant reduction in local tumour recurrence (OR 0.54, 95% CI 0.40–0.72, $p < 0.01$) was found in the CPT group. There was no statistical significance in the difference of mortality, enucleation rate, and cataract rate among the two groups. A statistically significant higher rate of secondary glaucoma (OR 1.81, 95% CI 1.24–2.63, $p < 0.01$) was found among those treated with CPT.

and provided an alternative way to confirm the robustness and validity of our findings.

DISCUSSION

The analysis of controlled studies revealed no significant differences in risks of mortality ($p = 0.57$) and enucleation ($p = 0.96$) among the CPT group and the brachytherapy group. Pooling analysis of case series that included latest studies using the random effects meta-regression model mentioned above showed similar results. In analysis of both controlled studies and

case series pooling, local recurrence rates ($p < 0.01$; $p < 0.01$) were significantly reduced in the CPT group.

The above mentioned findings were consistent with the first comprehensive systematic review and meta-analysis by Wang et al. [147] in 2013, which had reviewed 27 studies with 8809 uveal melanoma patients. The results showed no significant differences in mortality and enucleation rates but lower local recurrence rate treated with CPT. Statistics from the COMS trial were used as comparison arm, enrolling patients from 1985 to 2003. Treatment methods have improved since then, and recent studies revealed better outcome and fewer side effect [148]

Table 2. Meta-regression results of efficacy pooled from case series comparing CPT and brachytherapy.

		No. of patients	No. of studies included	OR	95% CI		p value
Death	CPT	17912	30				
	Brachytherapy	7574	36	1.32	0.99	1.77	0.06
Enucleation	CPT	21028	34				
	Brachytherapy	16092	78	1.16	0.87	1.54	0.31
Local tumour recurrence	CPT	19994	38				
	Brachytherapy	15432	82	0.54	0.4	0.72	<0.01
Cataract	CPT	6553	12				
	Brachytherapy	4404	44	1.59	0.87	2.93	0.13
Secondary glaucoma	CPT	11880	27				
	Brachytherapy	6195	41	1.81	1.24	2.63	<0.01

Table 3. Sensitivity test.

	OR	95% CI		p value
Death	1.38	0.53	3.58	0.4
Enucleation	0.85	0.24	3.03	0.71
Local recurrence	0.27	0.06	1.19	0.07

Sensitivity test was conducted by obtaining the odds ratio of outcomes between the CPT group and brachytherapy group in each controlled study and summarized the results using Mantel-Haenszel method.

compared to COMS trial. Furthermore, the use of iodine-125 therapy in the treatment of uveal melanoma has undergone modifications from 100 Gy to 80 Gy in 2003 to reduce collateral damage to surrounding tissues [29]. More recent studies which applied the revised treatment dose were not included in COMS trial or the review conducted in 2013.

By enrolling updated literature, we were able to include more relevant cases, achieve a more balanced distribution of patients between the two treatment groups, and provide a more comprehensive analysis of the efficacy and safety of brachytherapy in comparison to CPT for the treatment of uveal melanoma, which is a notable strength of our study. In addition, we conducted a comparative analysis between CPT and brachytherapy including head-to-head studies, which, to our knowledge, has not been reviewed and analyzed before.

Local control rate is an important issue concerning eye-preserving treatment of uveal melanoma because of high metastatic rate after local recurrence [149]. It is thus crucial to evaluate the recurrence rate of these two eye-preserving radiotherapies. Wang et al. reported significant reduction of local recurrence of uveal melanoma comparing CPT versus brachytherapy, with an OR of 0.22. Meanwhile, Buonanno et al. conducted a pooled analysis of 103 studies in 2022, which included 21,263 uveal melanoma patients treated with plaque brachytherapy, and reported a 91% local control rate after 5 years [150]. As for CPT, literature review of numerous reports showed a 5-year local control rate of 95% or greater [151].

Charged particles possess distinctive physical and biological characteristics, attributed to their positive charge and larger mass in comparison to photons or electrons. After penetrating tissue, the charged particles gradually decelerate and interact with neighbouring molecules. They are rapidly stopped over a short distance and release the majority of their energy, resulting in a sharp increase in dose at a specific depth known as the Bragg peak. The depth of Bragg peak is dependent on the initial kinetic energy of the particles. The particles preferentially damage targeted tumour cells, enhancing the local control rate while minimizing complications to adjacent tissue [11, 74, 152], which

might give an explanation to lower local recurrence rate in the CPT group in our analysis.

In our present study, analysis of case series studies revealed significantly higher incidence of secondary glaucoma after treatment with CPT compared with brachytherapy ($p < 0.01$). A similar trend was observed in the analysis of controlled studies, although no statistical significance was achieved ($p = 0.23$). It is important to note that charged-particle radiation therapy involves the administration of radiation to the anterior segment in order to reach posterior uveal melanoma tumours. As a result, a strong correlation has been established between the percentage of lens and anterior chamber involvement and the development of neovascular glaucoma [153, 154]. This adverse event is of significant importance, as neovascular glaucoma is recognized as the most important cause of secondary enucleation after radiotherapy [155]. Past research suggested that these complications could be reduced by reducing the dose to the anterior segment of the eye [154].

Due to the low incidence rate of uveal melanoma, conducting clinical trials is challenging. Moreover, there have been only a few charged-particle therapy facilities that treat uveal melanoma worldwide. These lead to some inevitable limitations in the present study. First, lack of literature with high quality and levels of evidence assessed by Cochrane risk of bias tools was observed. There is only one randomized controlled trial that has compared the efficacy of CPT and brachytherapy head-to-head [19], and its long-term result was included in our analysis. Even with the limited literature available, we have made effort to systematically review and enrol as many updated relevant studies as possible.

Secondly, there have been only a limited number of controlled trials on this topic, and the parameters used for outcome measurement were slightly different in each study. In order to conduct statistical analysis of important outcomes, we have simplified certain parameters. More specifically, reports on overall survival, disease-specific survival, overall mortality, and disease-specific survival were condensed and converted into a single measurement of mortality. The risks of enucleation, local recurrence, and ocular complications at various follow-up durations (e.g., 3 years, 5 years) were also merged into a single parameter. Notably, simplification of proton ion therapy with less commonly used carbon ions and helium ions in the CPT group may raise questions, given the limited global clinical utilization of the latter two [5]. In addition, simplification of ruthenium-106 and iodine-125 in brachytherapy group may also introduce heterogeneity since studies have suggested that ruthenium-106 may be less effective in treating thicker tumours [38]. While the simplification may contribute to the information bias and heterogeneity of our analysis, it was necessary in the absence of substantial evidence.

Thirdly, the relevant studies included differs in terms of study design, patient characteristics, treatment regimens, and outcome measures, contributing to inevitably high heterogeneity (I² index of >50% and heterogeneity *P* value of <0.05 in most of the outcome analysis). This may make it challenging to draw definitive conclusions and generalize the findings to a broader population. To reduce the influence of heterogeneity, random-effect size and meta-regression models were used for analyzing our results [156].

CONCLUSION

The findings of our study indicated that there was no significant difference in mortality, enucleation rate, and cataract formation rate between patients treated with CPT and brachytherapy. The CPT group showed lower local recurrence rate and a potentially higher incidence of secondary glaucoma. Nevertheless, the overall level of evidence is limited, and further high-quality studies are necessary to provide a more definitive conclusion.

SUMMARY

What was known before

- There was no significant evidence to determine the difference in mortality or enucleation rates comparing CPT group with brachytherapy group.
- Based on current evidence, local recurrence rate was significantly less in CPT than in brachytherapy.
- No systematic review and meta-analysis had compared the two treatment modalities head-to-head.

What this study adds

- We enrolled the most updated literature and thus achieving a more balanced distribution of patients between CPT and brachytherapy groups for meta-analysis.
- The present study included current head-to-head comparison studies of the two treatment modalities.

DATA AVAILABILITY

This study is registered was registered with PROSPERO (ID:399075). Extracted data are available online([link](#)); additional requests may be made to the corresponding author.

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AUTHOR CONTRIBUTIONS

YHT, CAH and YBC were responsible for designing the review protocol, screening potentially eligible studies, and interpreting results. YHT and CAH were responsible for conducting the search, extracting and analyzing the data, updating reference lists and drafting the original article. YBC supervised and provided feedback on the report. All authors have read and agreed to the published version of the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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Correspondence and requests for materials should be addressed to Yu-Bai Chou.

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